

TrES-5b

R-2405

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_140610 · created 2026-08-21T00:49:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456818.7776 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.044 +/- 0.036
Transit depth (Rp/Rs)^2	0.19 +/- 0.31 [%]
Orbital Inclination (inc)	89.36 +/- 3.13
Airmass coefficient 1 (a1)	1.419 +/- 0.041
Airmass coefficient 2 (a2)	-0.24 +/- 0.019
Scatter in the residuals of the lightcurve fit is	2.9 %
Best Comparison Star	None
Optimal Aperture	2.68
Optimal Annulus	16.98
Transit Duration (day)	0.0752 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.044 $\pm$ 0.036 vs 0.142 (deviation 2.72 σ)
Duration fit vs published	0.0752 d vs 0.0758 d (deviation 0.07 σ)
Mid-time O-C	-0.8 min
Depth SNR	1.2
Residual scatter	2.9 %

Light curve

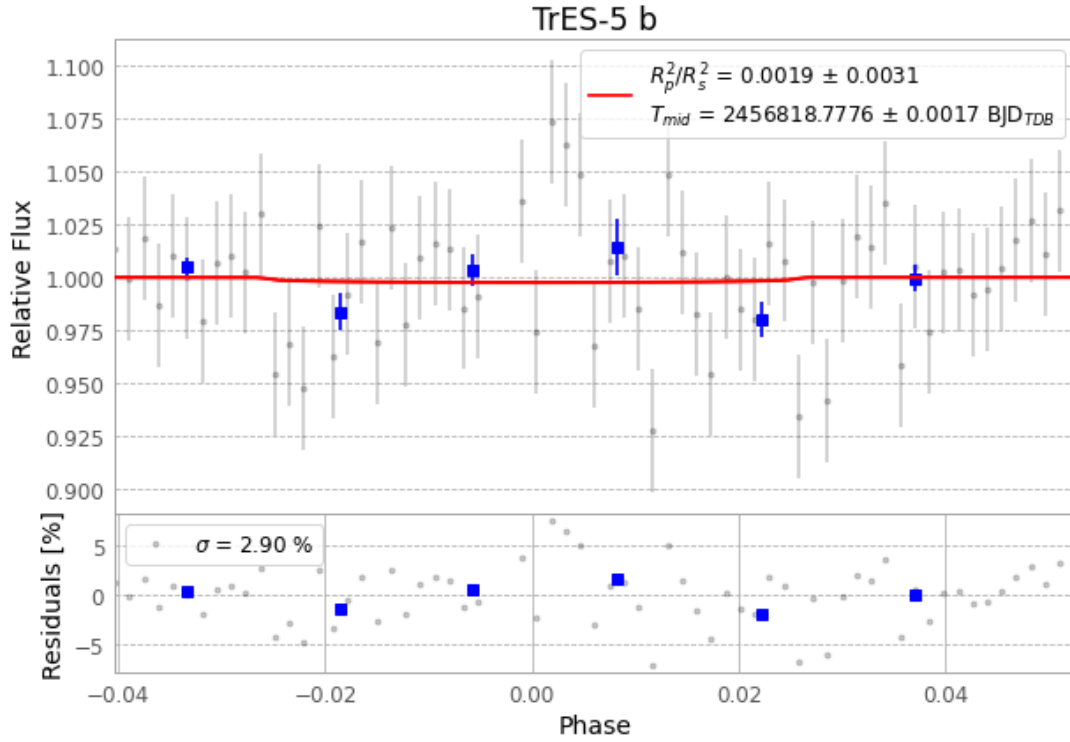


Figure 1 — FinalLightCurve_TrES-5 b_2014-06-09.png (SHA-256 9f9224ca0dae8a64...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [254, 247], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b62296a2c869adaa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-5140610051212.FITS → TRES-5140610083013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-140610.log (791bc4a1e8e2...), FinalLightCurve_TrES-5 b_2014-06-09.png (9f9224ca0dae...), FinalLightCurve_TrES-5 b_2014-06-09.csv (8a39231998ae...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:49Z.